# Loan Approval Prediction Case Study

# Internship Task 2 :- Data Science Project

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**Internship:** Alfido Tech Data Science Internship

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**Project Objective:**
The objective of this project is to build a machine learning model that predicts whether a loan application will be approved based on applicant characteristics such as income, education, employment status, credit history, and property area.

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**Dataset Description**

The dataset contains information about loan applicants and their loan approval status.

**Features**

- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area

### Target Variable

- Loan_Status
  - Y = Approved
  - N = Rejected

### Data Cleaning Steps

1. Removed Loan_ID column because it does not contribute to prediction.
2. Checked missing values using df.isnull().sum().
3. Filled missing numerical values using median.
4. Filled missing categorical values using mode.
5. Converted Dependents column from text to numeric.
6. Encoded categorical variables using Label Encoding.
7. Applied feature scaling using StandardScaler.
8. Handled class imbalance using SMOTE.

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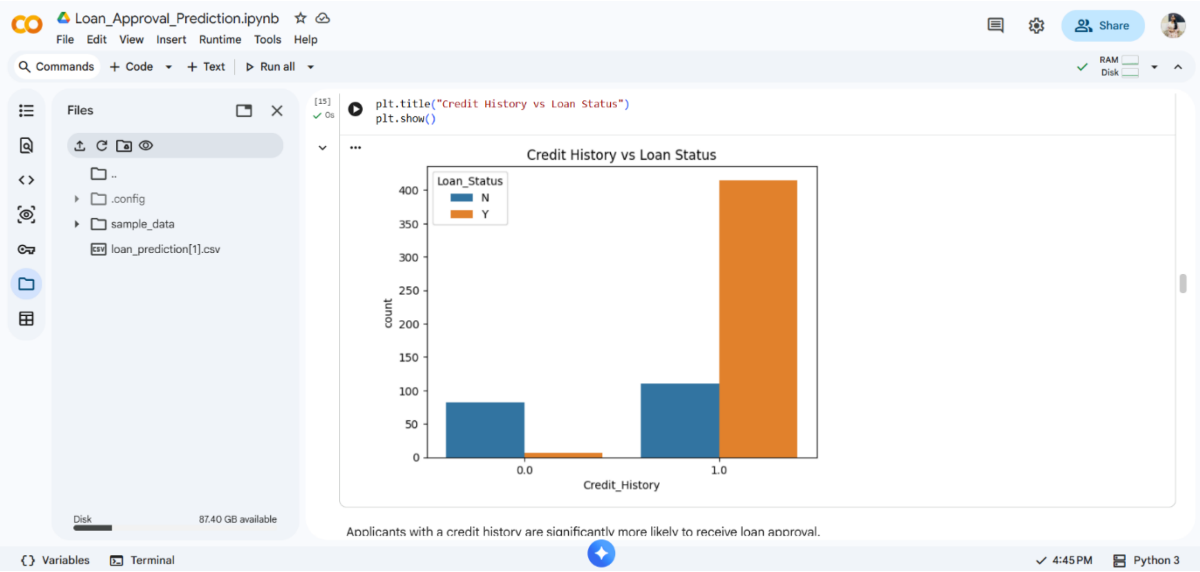
### Graph 1: Loan Approval Distribution


### Observation

Most applications were approved, indicating a moderate class imbalance.

## Graph 2: Credit History vs Loan Status




## Observation

Applicants with a valid credit history had significantly higher loan approval rates.

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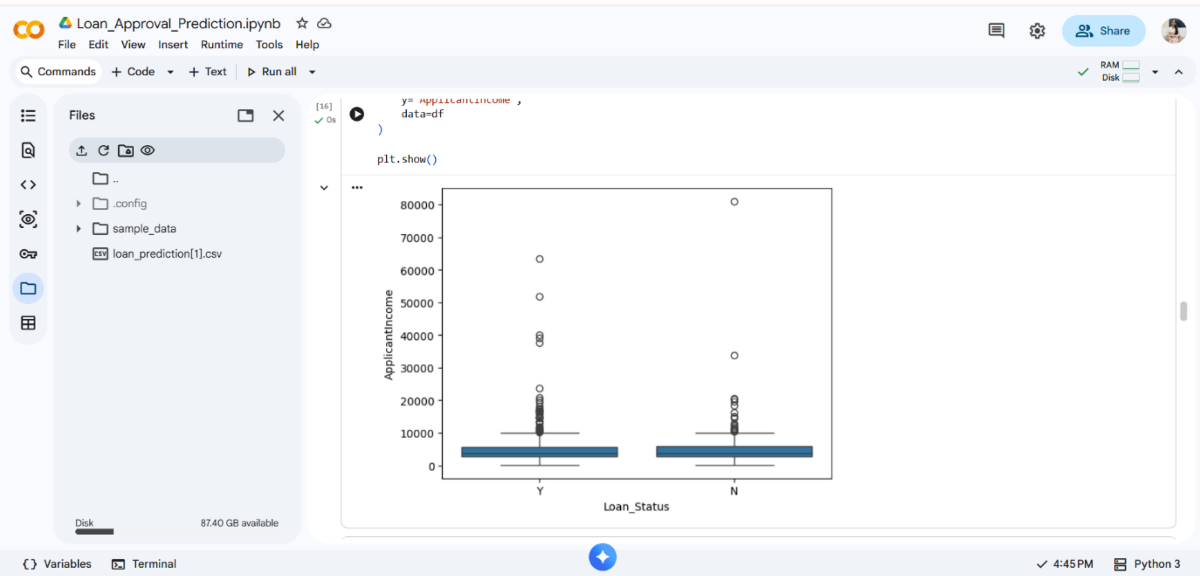
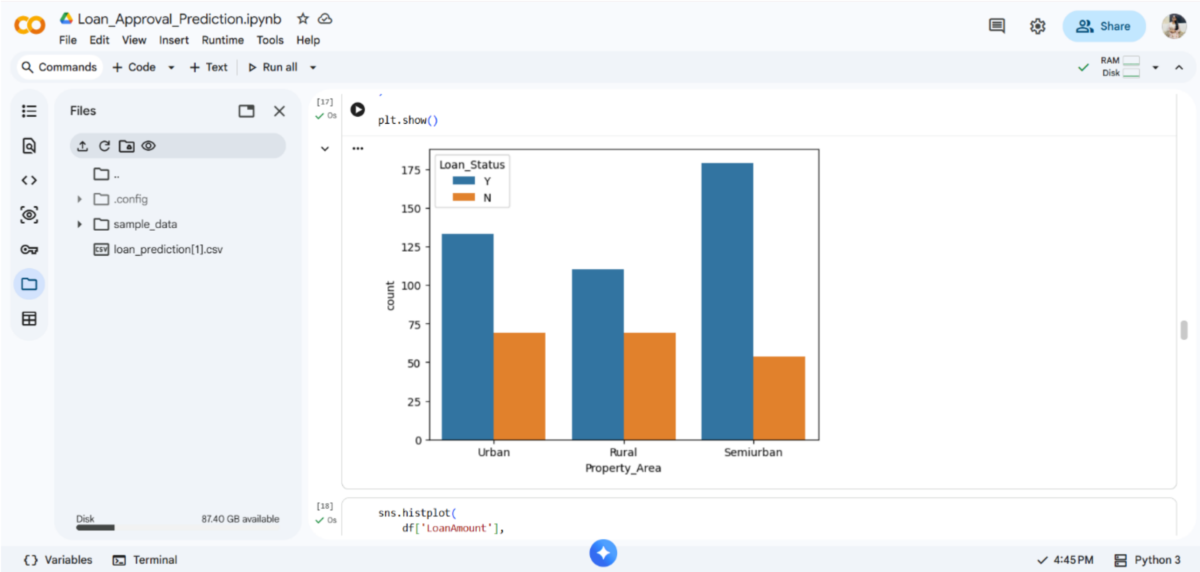
## Graph 3: Applicant Income Distribution




## Observation

Most applicants belong to lower and middle-income groups, while a few applicants have very high incomes.

## Graph 4: Property Area vs Loan Status




## Observation

Semiurban areas showed comparatively higher loan approval rates than urban and rural areas.

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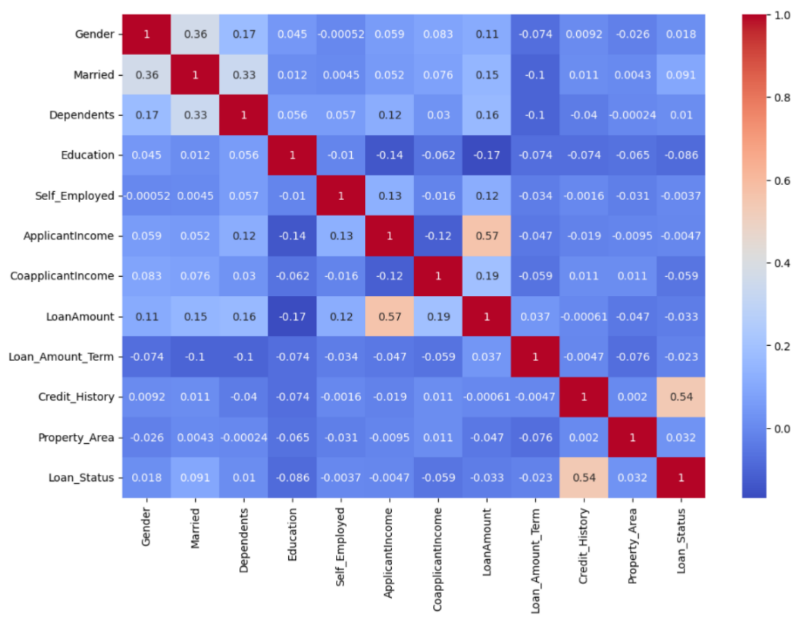
## Correlation Heatmap



**Key Findings**

1. Credit History is the most influential feature.
2. Applicant Income positively affects approval chances.
3. Loan Amount impacts loan approval decisions.
4. Property Area contributes moderately to prediction.
5. Education and employment status have smaller effects.

**Recommendations**

- Prioritize applicants with strong credit histories.
- Use predictive models to assist loan officers in decision-making.
- Regularly retrain the model using updated applicant data.
- Monitor model performance to ensure fairness and accuracy.

**Conclusion**

This project successfully developed a machine learning solution for loan approval prediction. After preprocessing, feature engineering, handling class imbalance, and comparing multiple models, Random Forest emerged as the best-performing model. The developed model can help financial institutions improve efficiency, reduce processing time, and support data-driven lending decisions.